



Year 11 Preliminary Mathematics Yearly Exam 2012

Name: _____

Tick your class/teacher	
11M1	
11M2	
11M3	
11M4	
11M5	

Examiner:

Time Allowed: 2 hours plus 5 minutes reading time

Start each question in a new booklet (5 booklets)

All questions are of equal value

You must show all necessary working

Full marks may not be awarded for careless or badly arranged work

Only calculators approved by the Board of Studies may be used

Include your name and the question number on the front of each booklet

All students to remain in the exam centre until the conclusion of the examination

	MC	Q1	Q2	Q3	Q4	Q5	Total	%
Total Marks	10	14	14	14	14	14	80	%

Answer these multiple choice questions on the sheet provided at the back

1 The correct factorisation of $8x^3 - 125$ is

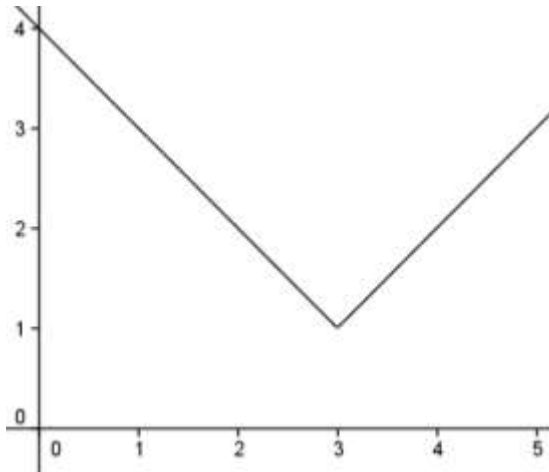
$$A \quad (2x-5)^3$$

B $(2x - 5)(4x^2 + 10x + 25)$

C $(2x+5)(4x^2-10x+25)$

D $8(x-5)(x^2-5x+25)$

2 The graph shown has equation



A $y = |x + 3| + 1$

B $y = |x - 3| + 1$

C $y = |x - 3| - 1$

D $y = |x + 3| - 1$

3 The value of $\sec 240^\circ$ is

A 2

$$B \quad \frac{2}{\sqrt{3}}$$

C -2

$$D \quad -\frac{2}{\sqrt{3}}$$

4 A function is defined for $x = a$. It is defined for other values of x . $\lim_{x \rightarrow a} f(x) \neq f(a)$
Therefore the function

A is continuous at $x = a$

B is differentiable at $x = a$

C is not continuous at $x = a$

D cannot be determined

5 In simplified form, the algebraic expression $\frac{x+5}{(x-3)(x+1)} - \frac{x-1}{x^2-x-2}$ can be written as

A $\frac{7x-13}{(x+1)(x-2)(x-3)}$

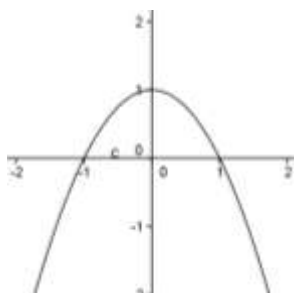
$$\text{B} \quad \frac{-x-13}{(x+1)(x-2)(x-3)}$$

$$\text{C} \quad \frac{7x-7}{(x+1)(x-2)(x-3)}$$

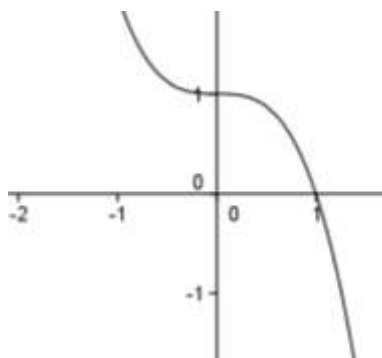
$$\text{D} \quad \frac{-x-7}{(x+1)(x-2)(x-3)}$$

- 6 A function possesses the property that $f(-x) = -f(x)$. Which of the following graphs could represent the function?

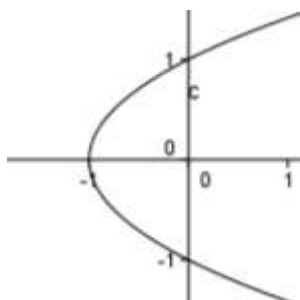
A



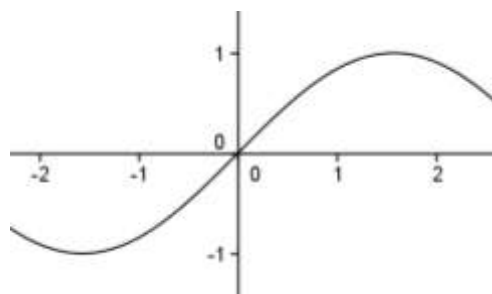
B



C



D



- 7 What is the equation for a circle of radius 9 units, which has its centre at $(-2, 5)$

A $(x-2)^2 + (y+5)^2 = 81$

B $(x+2)^2 + (y-5)^2 = 81$

C $(x-2)^2 + (y+5)^2 = 3$

D $(x+2)^2 + (y-5)^2 = 3$

- 8 Complete the following trigonometric identity $\operatorname{cosec}^2 x =$

A $1 + \cot^2 x$

B $\sec^2 x - 1$

C $1 - \tan^2 x$

D $\cos^2 x - \sin^2 x$

- 9 A triangle XYZ has $x = 14\text{cm}$, $y = 13\text{cm}$, $\angle Z = 105^\circ$. The side length z can be calculated using

A Cosine Rule

B Sine Rule

C cannot be calculated

D It is the ambiguous case

- 10 A triangle MNP has $m = 12\text{cm}$, $n = 14\text{cm}$, $p = 10.5\text{cm}$. Its smallest angle to the nearest degree is

A 46°

B 47°

C 39°

D 40°

Q11 Commence each answer in a fresh booklet**Marks**

Outcome P3 Performs arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities

a) Factorise:

(i) $x^2 - 11x$ **1**

(ii) $3x^2 - 4x - 7$ **2**

(iii) $x^3 + 8$ **2**

(b) Solve:

(i) $\frac{8}{x} + 2 = \frac{16}{x}$ **2**

(ii) $|2x - 6| \leq 4$ **3**

Graph your solution on the number line

(c) Solve the simultaneous equations: **2**

$xy + y^2 = 54$

$x + y = 9$

(d) Find $0.7\dot{1}$ as a fraction in simplest terms **2****Q12 Begin a fresh booklet for Q2**

Outcome P3 Performs arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities

(a) Find the value of **1**
 $27^{\frac{1}{3}} + 2^{-2}$ (b) Give the value of $1.79^{1.35}$ correct to 3 decimal places **1**(c) Simplify the following $2\sqrt{20} + \sqrt{45}$ **2**(d) Rationalise the denominator and simplify fully **2**
$$\frac{4 + \sqrt{2}}{\sqrt{5} + \sqrt{3}}$$
(e) Expand and simplify $(3 + 2\sqrt{5})(2 - \sqrt{5})$ **2**(f) Find values for a and b by completing the square **2**
 $x^2 - 8x + 4 = (x - a)^2 - b$

(g) Find the following trigonometric ratios in exact terms

(i) $\cos 240^\circ$ **1**

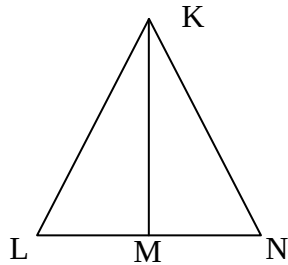
(ii) $\cot 330^\circ$ **1**

(h) Show that $\cos^2(90 - \theta) \cot \theta = \sin \theta \cos \theta$

2

Q13 Begin a fresh booklet for Q13

Outcome P4 Chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques



(a) The diagram shows an isosceles triangle $\triangle KLN$
With $KL = KN$. KM is constructed to bisect LN .

(i) Complete the proof that $\triangle KMN \equiv \triangle KML$ 3

KM common

$KN = KL$ _____

_____ = _____ by construction

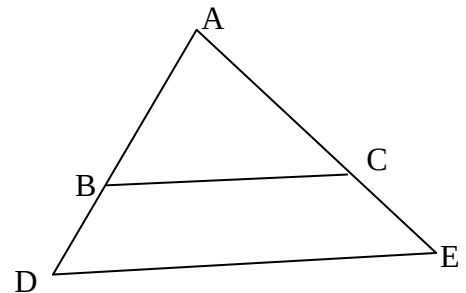
$\triangle KMN \equiv \triangle KML$ _____

(ii) Complete the proof that an isosceles triangle 1
has base angles equal

(b) $BC \parallel DE$.

(i) Prove $\triangle ABC \parallel \triangle ADE$ 3

(ii) $AD = 7$, $BD = 3$, $CE = 4.5$. Find AC 1



(c) A tree is situated on a level field. 3

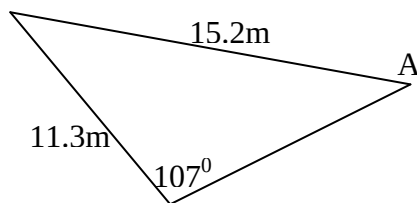
The sun is at an angle of elevation of 31° . The length of the shadow is 47m.

Draw a diagram to represent this information.

Calculate the height of the tree.

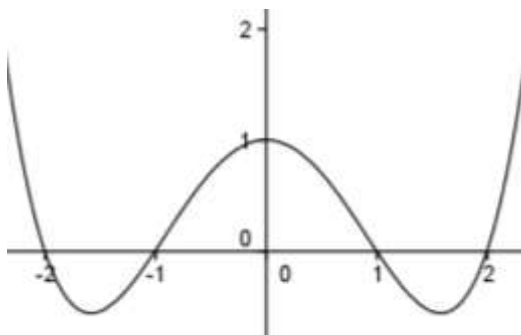
(d) Calculate the size of the angle A 3

to the nearest whole degree.



Q14 Begin a fresh booklet for Q14*Outcome P5 Understands the concept of a function and the relationship between a function and its graph*

- (a) What is the natural domain and the range for $y = \sqrt{9 - x^2}$ 3



- (b) The accompanying graph shows 3

a function. $y = \frac{x^4 - 5x^2 + 4}{4}$

Is it odd, even or neither?

Give both a graphical and an algebraic reason for your decision.

- (c) (i) Graph the function $y = \frac{6}{x}$ include at least 1 point. 2

- (ii) Is the function continuous? Be specific. 1

- (d) Shade the region specified by 2

$x^2 + y^2 \leq 25$ **or** $y \geq x$ on the number plane

- (e) (i) Sketch the relation $x = y^2$ 1

- (ii) Is this relation a function? You must give a reason for your decision. 2

You may make marks on your sketch to help with your justification.

Q15 Begin a fresh booklet for Q15*Outcome P6 Relates the derivative of a function and its graph*

(a) $\frac{dy}{dx} = 3 - 2x$

- (i) Find $f'(3)$ 1

- (ii) For what value of x does the tangent have a gradient of 5 1

- (b) Find $f'(x)$ for each of the following

(i) $f(x) = x^3 - 2x$ 1

(ii) $y = 2x - \frac{3}{x}$ 1

(iii) $y = (x^2 - 3x)^{-2}$ 2

- (c) Find the equation of the tangent to $y = x^2 - 3\sqrt{x}$ at the point where $x = 1$ 3

- (d) (i) Obtain the equation of the normal to $y = x^2 - 3$ at the point (2, 1) 2
- (ii) Where does this line cut the y axis? 1
- (e) Obtain the coordinates of the point on $y = x^2 - 3x + 2$ 2
where the tangent is parallel to the x axis.

☺ END OF PAPER ☺

Multiple Choice Answer Sheet

Name: _____

Teacher's Name _____

[illegible]

Prelim Exam (20) Mathematics

2012

1 B 2 B 3 C

4 C 5 A 6 D.

7 B. 8 A. 9. A.

10. B.

1 each.

/10

11 a) i) $x(x-11)$

1

ii) $(3x-7)(x+1)$

2

1 for program

iii) $(x+2)(x^2-2x+4)$

2

1 for program

b) i) $\frac{8}{x} + 2 = \frac{16}{x}$

$8+2x = 16$

$x = 4$

2

1 for program

ii) $|2x-6| \leq 4$

$-4 \leq 2x-6 \leq 4$

$2 \leq 2x \leq 10$

$1 \leq x \leq 5$

1

1

1.



c) $xy + y^2 = 54$ ----- ①

$x+y = 9$ ----- ②

P2 SOLS Prelim Final Yr 11 MATHS P2

11 c) contd,

from ② $y = 9 - x$. sub into ①.

$$x(9-x) + (9-x)^2 = 54.$$

$$9x - x^2 + 81 - 18x + x^2 = 54.$$

$$81 - 9x = 54.$$

$$27 = 9x$$

$$3 = x$$

$$y = 6.$$

(1 for progress)

d) $x = 0.71$

$10x = 7.11$

2

1

$$9x = 6.4.$$

$$x = \frac{64}{90} = \frac{32}{45}.$$

1

14

12 a) $3 + \frac{1}{4} = 3.25.$

1

b) 2.195

1

c) $\frac{2 \times 2\sqrt{5} + 3\sqrt{5}}{7\sqrt{5}}.$

1

1

d) $\frac{4+\sqrt{2}}{\sqrt{5}+\sqrt{3}} \cdot \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}-\sqrt{3})}$

1

$$\frac{4\sqrt{5} - 4\sqrt{3} + \sqrt{10} - \sqrt{6}}{2}.$$

1

e) $(3+2\sqrt{5})(2-\sqrt{5}) = 6 - 10 - 3\sqrt{5} + 4\sqrt{5}$
 $= \sqrt{5} - 4.$

1

1

f) $x^2 - 8x + 4 = x^2 - 8x + 16 - 12$

$= (x-4)^2 - 12$ $a=4$
 $b=12$

g) $\cos 240 = -\cos 60 = -\frac{1}{2}$

$\cot 330 = \frac{-1}{\tan 30} = -\sqrt{3}$

h) $\cos^2(90-\theta) \cot \theta$
 $= \sin^2 \theta \times \frac{\cos \theta}{\sin \theta}$

$= \sin \theta \cos \theta$ as req^d

14

Q13. given

$LM = MN$

SSS rule

$LL = LN$ corresponding Ls in
congruent Δ s.

$\angle ABL = \angle ADE$ corresponding Ls
parallel lines.

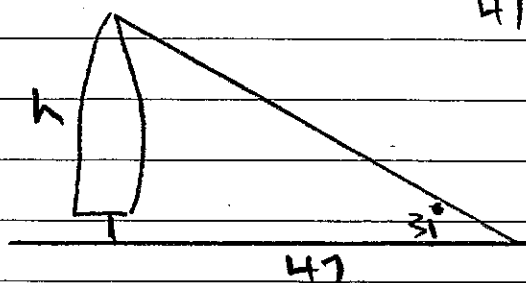
$\angle A$ common (or $\angle ACP = \angle AED$)
corr^y // lines.

$\Delta ABL \parallel \Delta ADE$ equiangular

ii) b

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sun



$$\frac{h}{47} = \tan 31$$

$$h = 47 \tan 31$$

$$= 28.24 \text{ m.}$$

d) $\frac{\sin A}{11.3} = \frac{\sin 107}{15.2}$

$$\sin A = 0.7109 \dots$$

$$A = 45 \text{ or } 135$$

but 135 would be too big

$$\therefore A = 45$$

Must be
sure about
obtuse.

14

14 a) Domain $-3 \leq x \leq 3$

Range $0 \leq y \leq 3$

2 allow 1 for
 $x \leq 3$

b) even

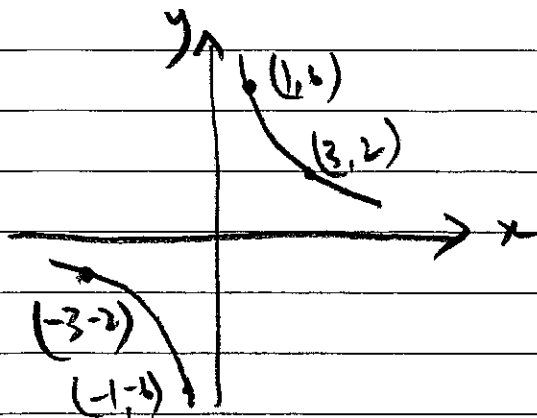
Symmetric about y axis.

$$f(-x) = x^4 - 5x^2 + 4 = f(x)$$

c) i) PTO

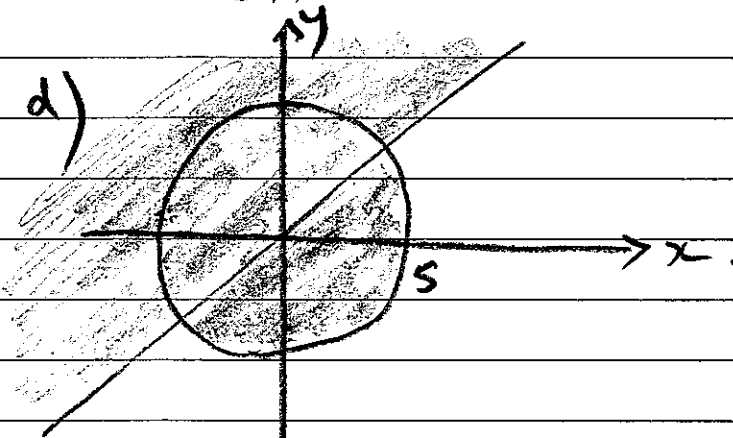
ii) Not continuous when
 $x = 0$

c) i)



2
must have ≥ 1
point specified

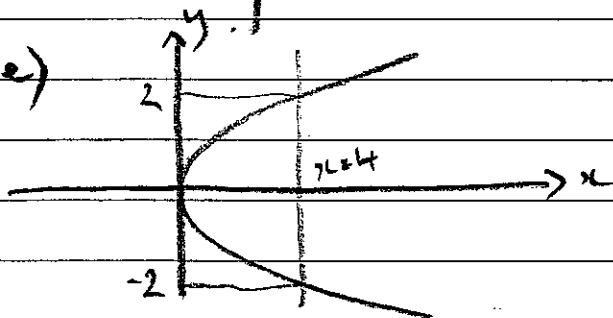
d)



1 for $0+$ ✓

1 for shading.

e)



ii) Not function.
for example if $x=4$ $y=\pm 2$

15. $f'(3) = 3 - 2 \times 3 = -2$

a)

$$5 = 3 - 2x$$

$$2x = -2$$

$$x = -1$$

b). 1) $f'(x) = 3x^2 - 2$

1

1

15b) ii) $f(x) = 2 + 3x^{-2}$
 $= 2 + \frac{3}{x^2}$

iii) $f'(x) = -2(x^2 - 3x)^{-3}$
 $\times 2x - 3$

$= \frac{-4x + 6}{(x^2 - 3x)^3}$

no need to render in
simplified form
(-1) if error made

c) $y = x^2 - 3x^{\frac{1}{2}}$
 $y' = 2x - \frac{1}{2}x^{-\frac{1}{2}}$

$f(1) = 2 - \frac{1}{2} = \frac{1}{2}$ M.F.

$y_1 = 1 - 3 = -2$

$y + 2 = \frac{1}{2}(x - 1)$

$y = \frac{1}{2}x - 2\frac{1}{2}$

d) i) $y' = 2x$ $f'(2) = 4$
 $M_N = -\frac{1}{4}$

$y - 1 = -\frac{1}{4}(x - 2)$

$4y - 4 = -x + 2$ / OR $y = -\frac{1}{4}x + \frac{3}{2}$
 $x + 4y - 6 = 0$

1 must be simplified
into a form.

y axis $x = 0$ $y = \frac{3}{2}$

e) $y' = 2x - 3$ if $y' = 0$ $x = \frac{3}{2}$
 $f(\frac{3}{2}) = -\frac{1}{4}$ $(\frac{3}{2}, -\frac{1}{4})$